

BIOL 343

Applied Bioinformatics

Intro to RNA sequencing

Dr. Nic Wheeler

Introduction



Class introductions

Name (and pronouns if desired), year of study, major

Why you took this class

One thing you hope to learn in this class

One thing you like about yourself



Course introduction

Welcome to the *University of Wisconsin-Eau Claire*
Bioinformatics Core Facility!

Links: GitHub organization profile and BIOL 343 repository



Weekly agenda

Lecture Monday

Content delivery on a particular aspect of RNA sequencing

Tool Tuesday

Introduction of RNA-seq bioinformatics tools using example data and Jupyter notebook

Dry Lab Thursday

In-class homework using Tuesday's tools on data selected by the student

Semester agenda

Weeks 1-8

RNA-seq bootcamp on
published data

Weeks 9-14

Independent project on novel
data

Syllabus quick hitters

- 1. Grading
- 2. Drop-in hours
- 3. AI

Category	A	B	C	D	F
Attendance	>=13/14 perfect weeks	12/14 perfect weeks	11/14 perfect weeks	10/14 perfect weeks	Less than 10/14 perfect weeks
Homework	>=8/9 satisfactory homework submissions	7/9 satisfactory homework submissions	6/9 satisfactory homework submissions	5/9 satisfactory homework submissions	Less than 5 satisfactory homework submissions
Assessments	Requirements for (B), plus one of two tracks: <i>a working snakemake pipeline (Bioinformatics)</i> or <i>a ≥90% on the content exam (BMB)</i>	Requirements for (C), plus version-controlled work	Complete the full differential expression pipeline in a notebook (rubric-satisfying) and pass the content exam (≥70%)	Partial progress through the pipeline	Unsatisfactory progress through the pipeline

Lectio - your BIOL 343 AI assistant

Lectio introduction

Lectio

Applied Bioinformatics

New Chat

Quiz Me

Nic Wheeler

Explain why RNA quantification is important to biology research

Resend

Lectio

Why RNA Quantification is Important to Biology Research

mRNA quantity is indicative of cellular function or status[1]. This makes RNA quantification a powerful tool for understanding what's happening inside cells.

The Connection Between mRNA and Protein Function

mRNA quantity is correlated with protein quantity[2], which is crucial because proteins are the primary drivers of cellular function. By measuring mRNA levels, researchers can infer which proteins are being produced and at what levels, giving them insight into the cell's current state and activities.

Regulatory Insights

Differential protein expression determines cellular function[2]. Understanding mRNA expression allows researchers to identify which genes are "turned on" or "turned off" in response to different conditions, treatments, or disease states. This reveals how cells regulate their behavior at the molecular level.

Practical Applications

Consider a simple example: if you measure mRNA from genes involved in metabolism, cell division, and immunity, the relative expression levels of these genes tell you what the cell is prioritizing. High metabolism gene expression might indicate active energy production, while high immunity gene expression might suggest the cell is responding to a threat.

Intro To Rna Seq lecture slides

Page 7 of 21

Fit

Why quantify mRNA expression?
mRNA quantity is correlated with protein quantity

Differential protein expression determines cellular function

Four opportunities for regulation

1. Transcription of mRNA

2. Degradation of mRNA

3. Translation of protein

4. Degradation of protein

Central Dogma

Transcription

Processing

Translation

DNA

Pre mRNA

Mature mRNA

Peptide

Rates of:

Transcription

mRNA translation

Protein degradation

mRNA degradation

73%

8%

8%

11%

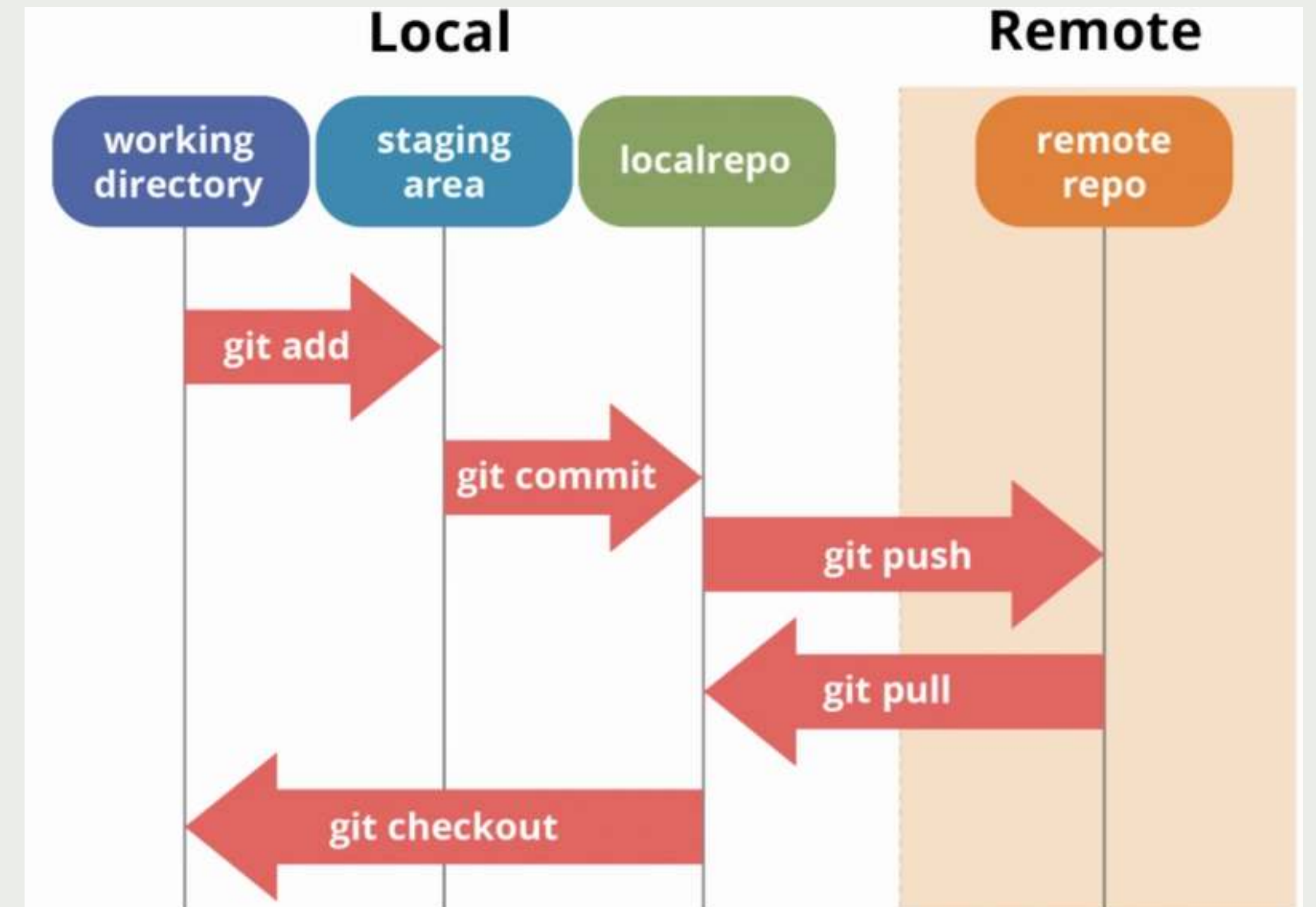
Ask a question about the learning objectives...

1. Visit lectio.study
2. Login with your UWEC credentials

GitHub and version control

Terminology

- Working directory
- Staging area
- Repository (repo)
- Commit
- Local
- Remote
- Checkout/clone
- Push/pull



GitHub and version control

1. Make GitHub profile
2. Join UWEC Bioinformatics organization
3. Setup SSH key for repo cloning
 1. Generate key on BOSE (no passphrase)
 2. Add key to GitHub account
4. Clone BIOL343 repo into BOSE home dir
5. Open up these slides

GitHub and version control

1. Move into repo: `cd BIOL343`
2. Make a new branch with your name: `git checkout -b {name}`
3. Make a new file: `nano temp.txt`
4. Edit the file
5. Stage: `git add .`
6. Commit: `git commit -m "{message}"`
7. Push: `git push`
8. Switch to main: `git checkout main`
9. Pull: `git pull`
10. Switch to your branch: `git checkout {name}`
11. Merge: `git merge main`

Questions?

Break if time...

Learning Objectives

You will be able to:

1. Explain why RNA quantification is important to biology research
2. Identify real-world uses of RNA sequencing in biology research and biomedical applications and explain in detail the use-case for the Hartig et al. 2016 paper
3. Describe the molecular biology behind historical approaches to RNA quantification and list their advantages/disadvantage(s)

Applied bioinformatics

Using sequences to tell us something about biology

Bioinformatics is the computational analysis of sequence data

- DNA
 - Species genetic differences
 - Population/individual genetic differences
 - Causes of genetic diseases
 - ***Quality (the actual sequence and variation in it)***
- RNA (typically mRNA)
 - Cellular function
 - Organismal/disease status
 - ***Quantity***

Write down everything you know about DNA structure and function...

Write down everything you know about RNA structure and function...

Types of RNA

mRNA

Messenger

Will be translated to protein

Alternatively spliced

Regulated (degradation)

Highly dynamic quantities

rRNA

Ribosomal

Never translated

Creates ribosomes (along
with proteins)

Highly stable

Highly expressed

tRNA

Transfer

Never translated

Binds to amino acids

Recognizes codons during
translation

Highly stable

Lowly expressed

Small RNA

Many types (miRNA,
snRNA, snoRNA, piRNA,
etc.)

Often regulate mRNA levels

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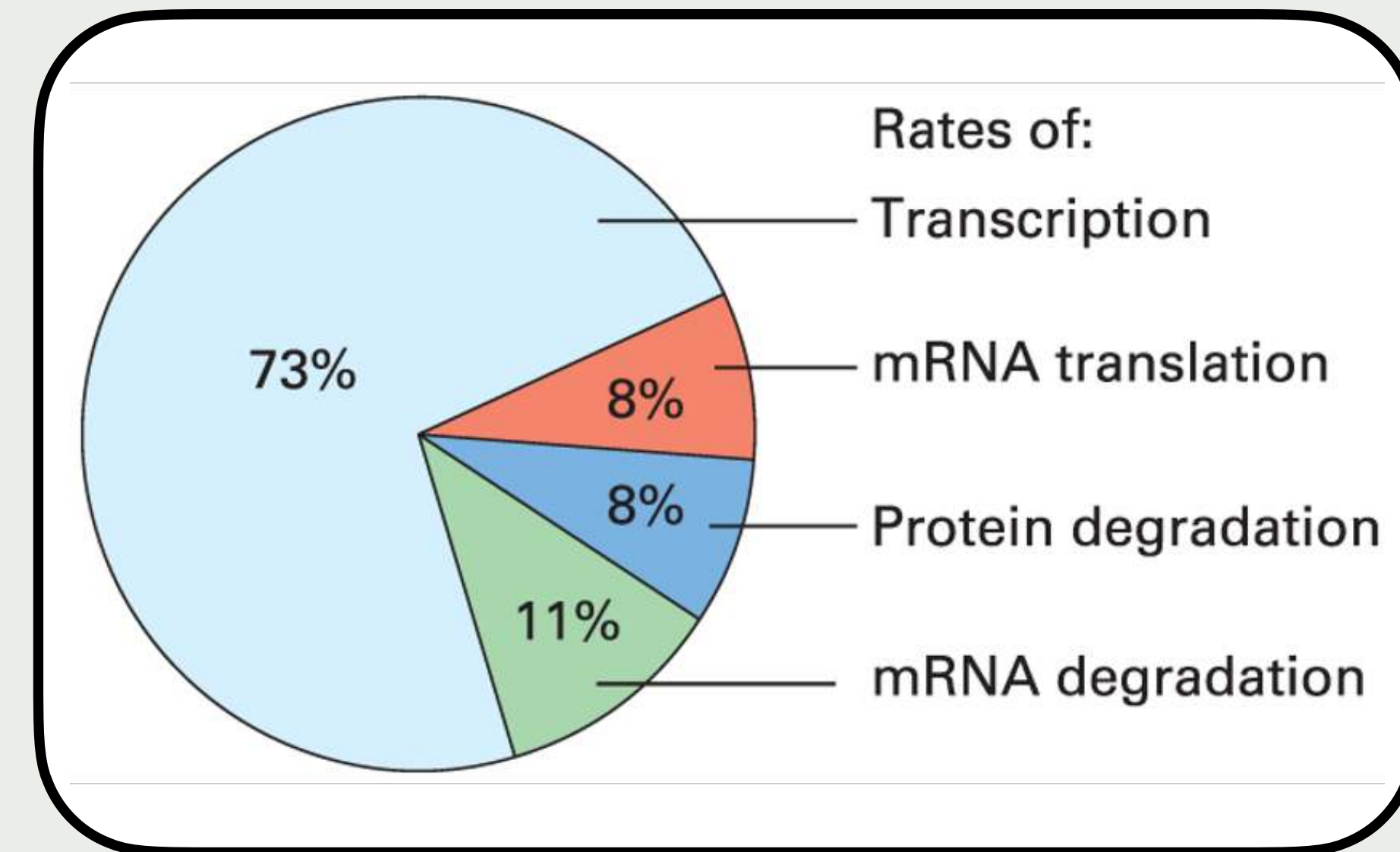
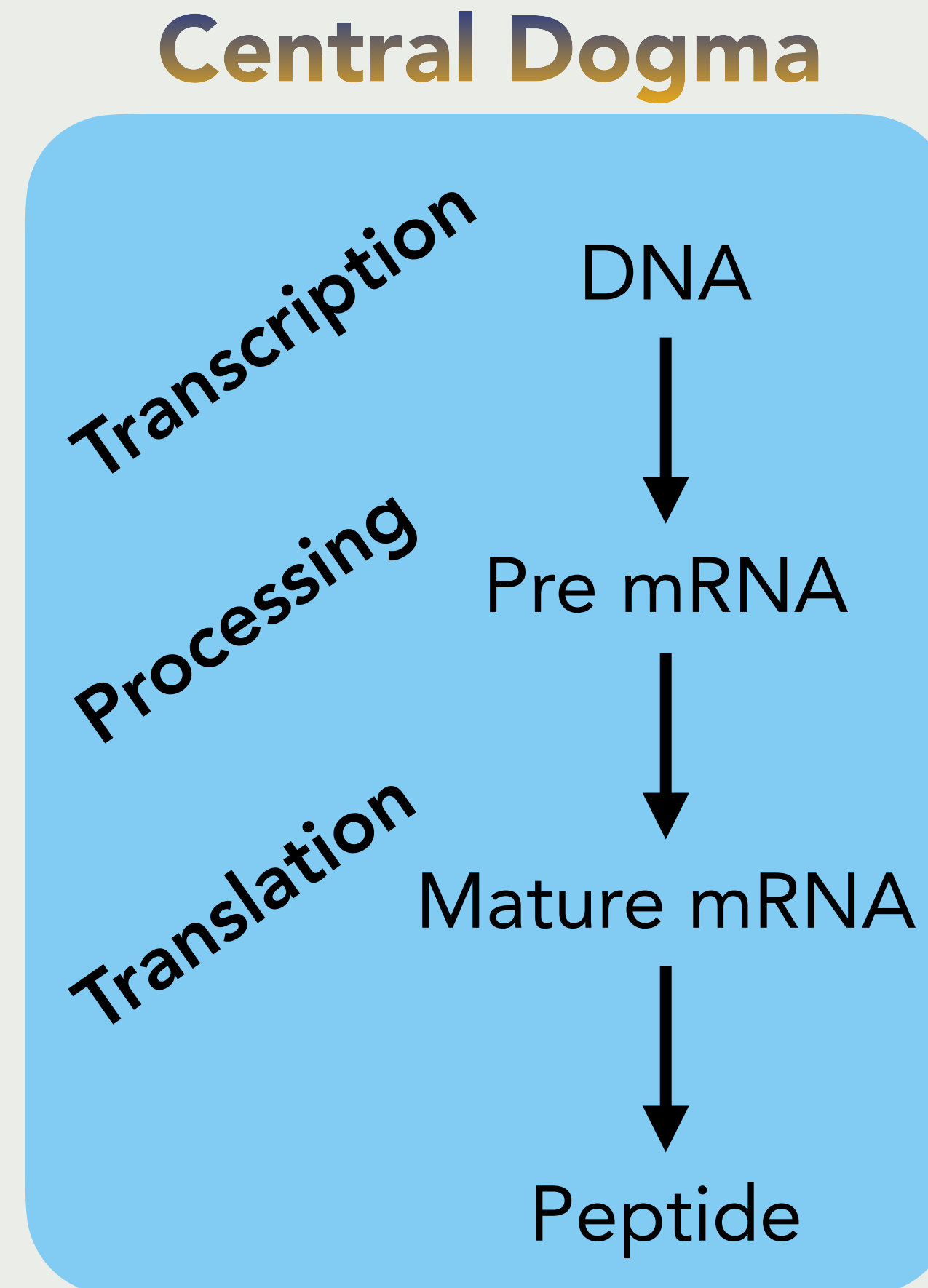
Often regulate mRNA levels

Sometimes...

Why quantify mRNA expression?

mRNA quantity is correlated with protein quantity

- Differential protein expression determines cellular function
- Four opportunities for regulation
 1. Transcription of mRNA
 2. Degradation of mRNA
 3. Translation of protein
 4. Degradation of protein



Why quantify mRNA expression?

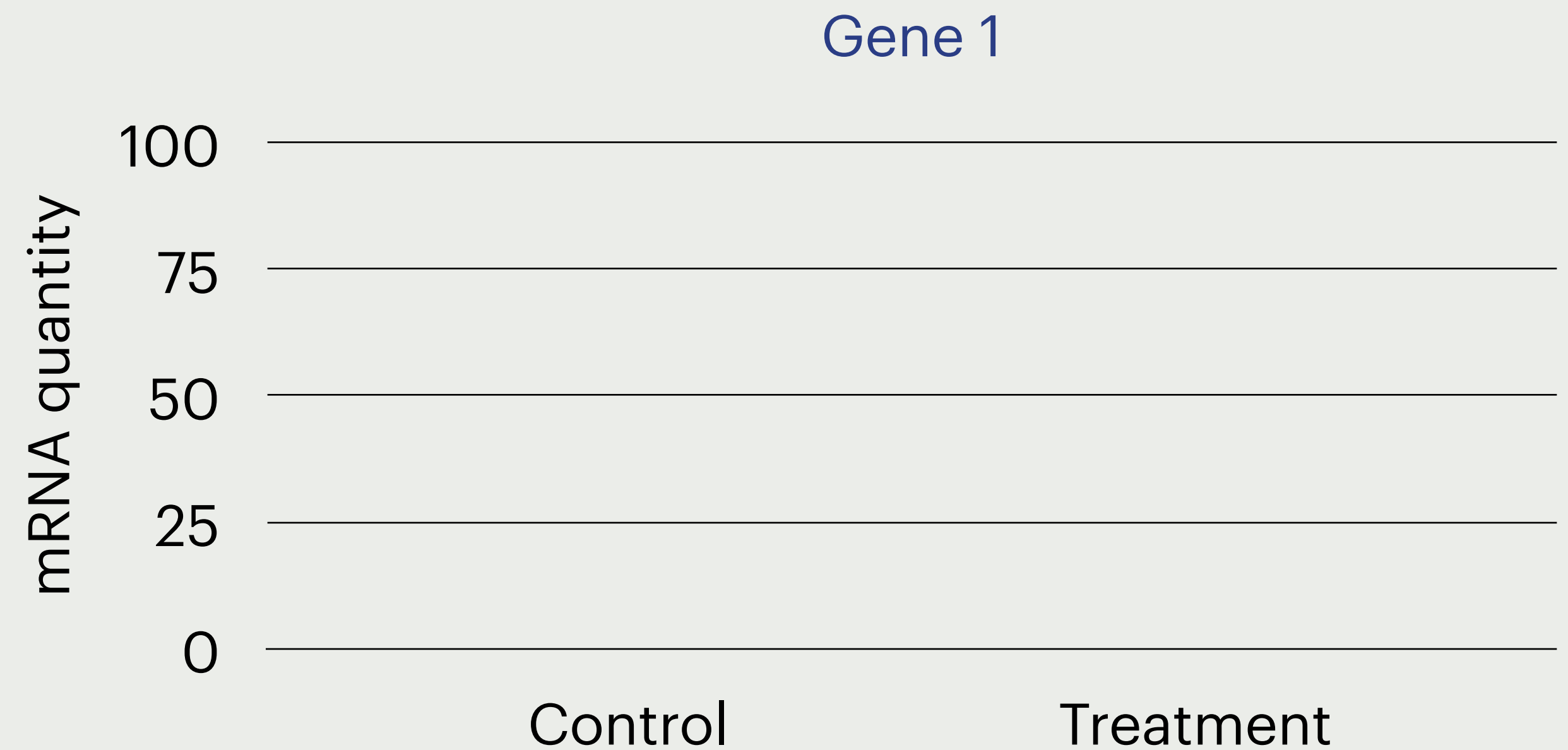
mRNA quantity is indicative of cellular function or status

What does the following data tell you about the function and status of the cell?

For example...

Suppose an animal has 1 protein-coding gene:

Gene 1 - living



Why quantify mRNA expression?

mRNA quantity is indicative of cellular function or status

What does the following data tell you about the function and status of the cell?

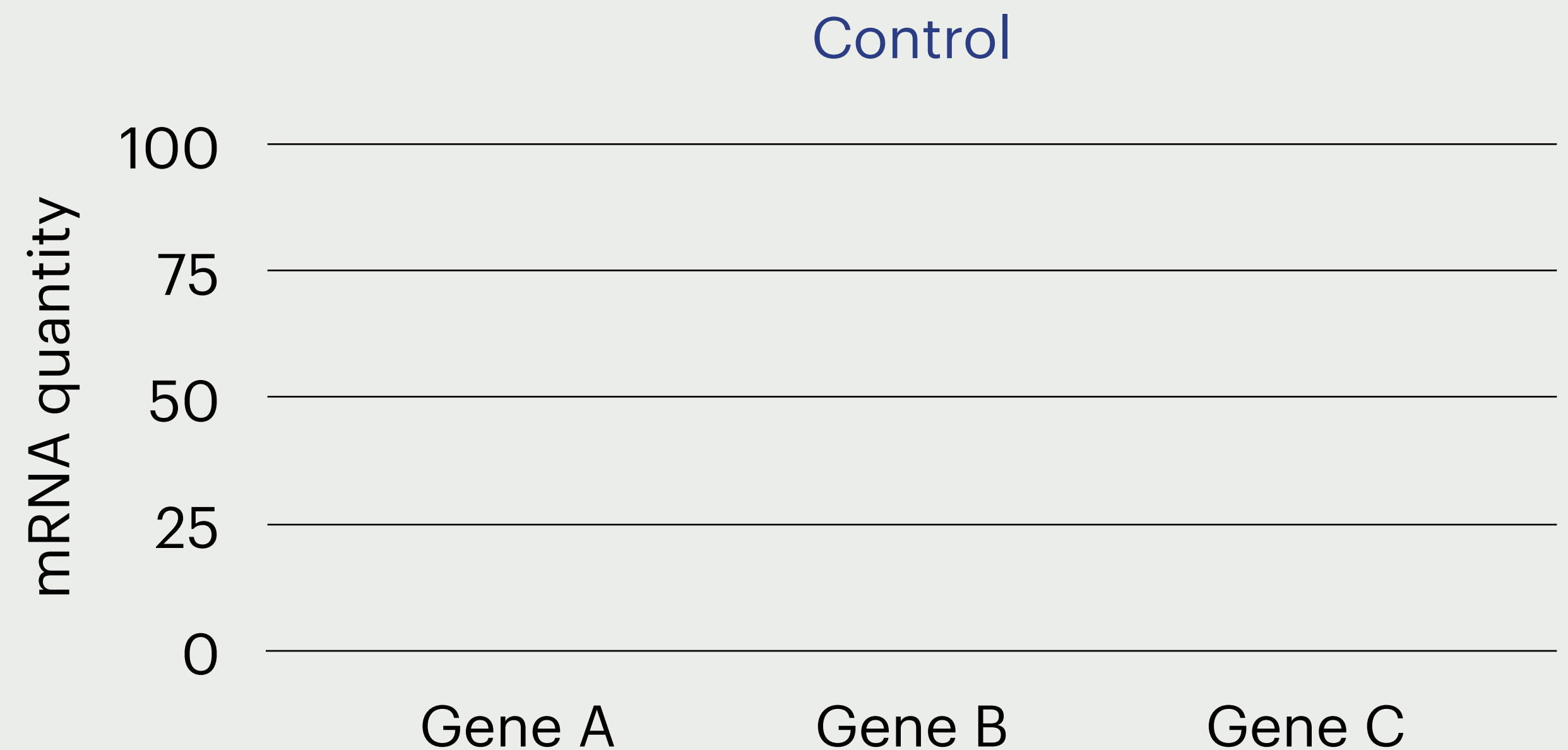
For example...

Suppose an animal has 3 protein-coding genes:

Gene A - metabolism

Gene B - cell division

Gene C - immunity



Why quantify mRNA expression?

mRNA quantity is indicative of cellular function or status

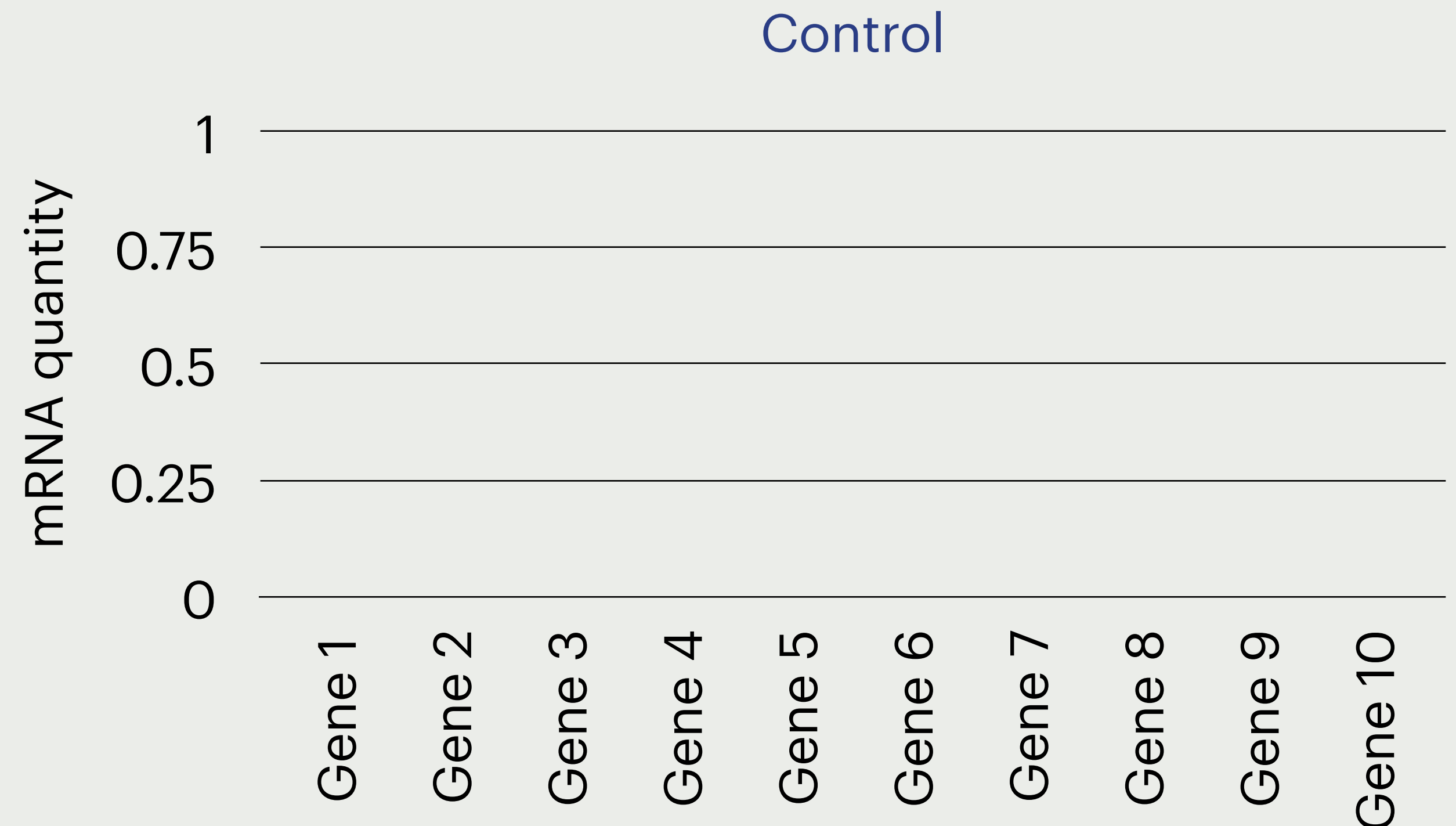
What does the following data tell you about the function and status of the cell?

For example...

Suppose an animal has 10 protein-coding genes:

Genes 1-5 - happiness

Genes 6-10 - sadness



Why quantify mRNA expression?

mRNA quantity is indicative of cellular function or status

For example...

Suppose an animal
protein-coding g

Genes 1-5 - ha

Genes 6-10 - sa

**What about when an
animal has 15,000
protein-coding genes?**

Following data tell
tion and status
ell?

control

mfold
0.25
0

Gene 1

Gene 2

Gene 3

Gene 4

Gene 5

Gene 6

Gene 7

Gene 8

Gene 9

Gene 10

Global analysis of RNA expression: *transcriptomics*

Transcriptome analysis is a powerful tool for comparing molecular biology

- Real-world examples of transcriptomics use-cases:
 - Identify genes that are highly/lowly expressed in metastatic tumors in comparison to non-metastatic tumors or healthy tissue
 - Identify marker genes for a disease state (i.e., Alzheimer)
 - Analyze changes in gene expression during embryonic development or wound healing
 - Prioritize genes expressed in the etiological agents of infectious diseases for study as potential druggable targets
 - Compare/contrast expression differences between different but related populations of animals
 - *What else??*

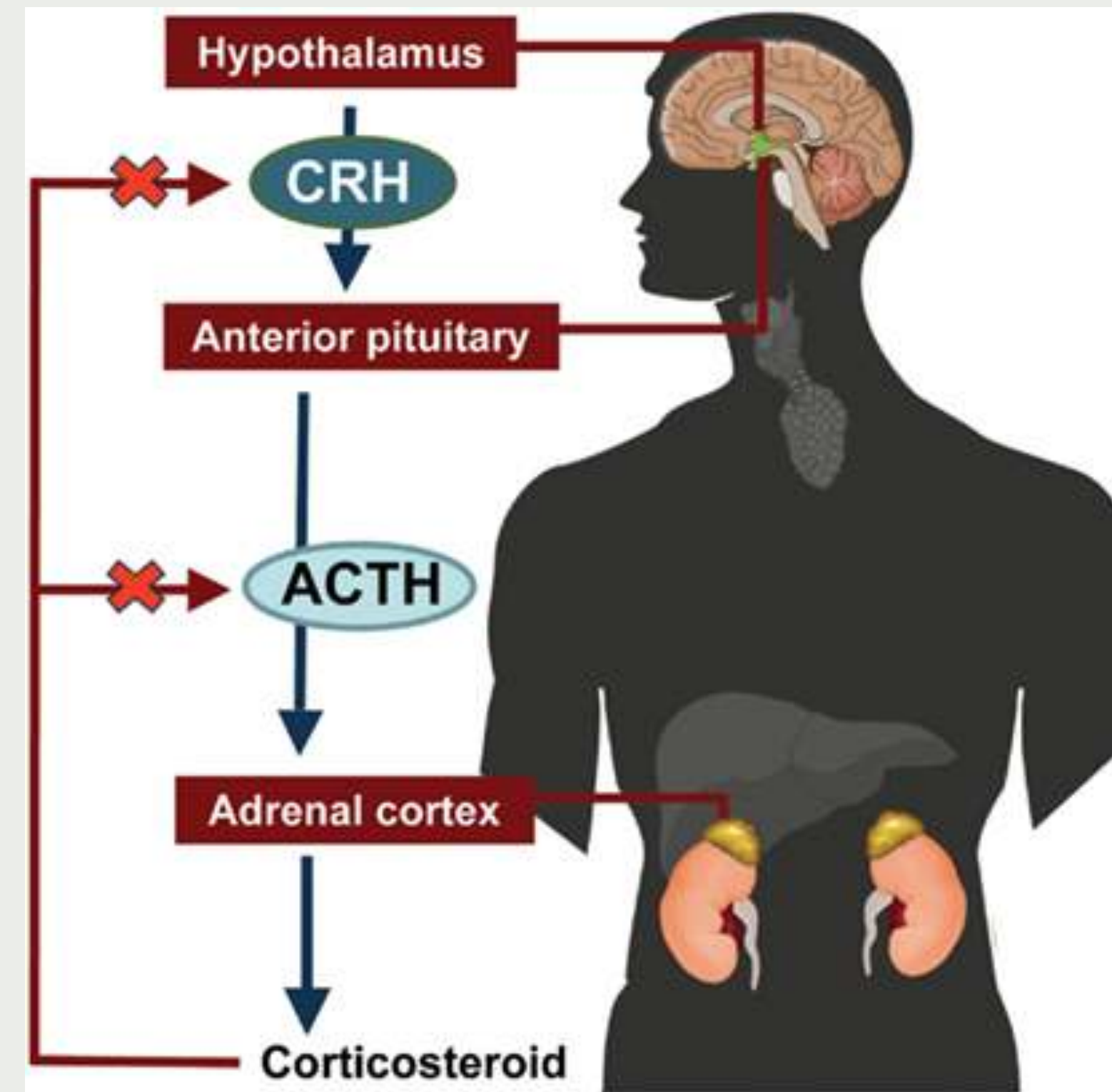
Generic goal: identify ***differentially expressed genes (DEGs)***

Why quantify mRNA expression?

Cortisol-treated zebrafish embryos develop into pro-inflammatory adults with aberrant immune gene regulation

Cortisol

- Primary mediator of the systemic stress response
- Release controlled by the hypothalamus-pituitary-adrenal (HPA) axis
 1. Stress signals in brain tell H to release corticotropin-releasing hormone (CRH)
 2. CRH stimulates the P to release adrenocorticotrophic hormone (ACTH)
 3. ACTH stimulates the A to release cortisol

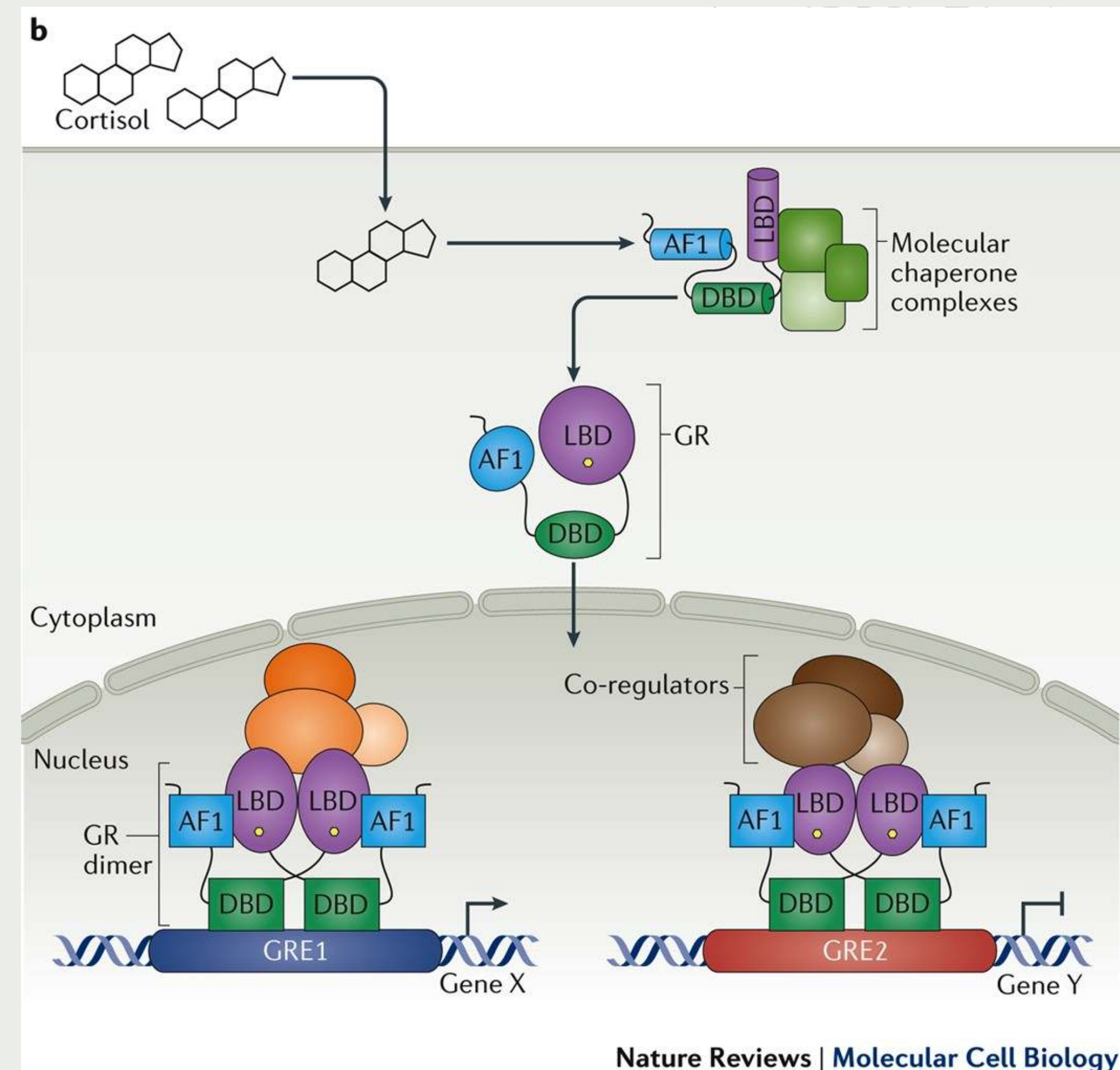


Why quantify mRNA expression?

Cortisol-treated zebrafish embryos develop into pro-inflammatory adults with aberrant immune gene regulation

Cortisol

- Cortisol is a hydrophobic steroid and can cross the plasma membrane
- In the cytoplasm, it binds to the glucocorticoid receptor (GR)
- GR dimers translocate to the nucleus to act as a transcription factor and bind to glucocorticoid response elements (GREs) in gene promoters
- GREs are present in genes that control metabolism and inflammation



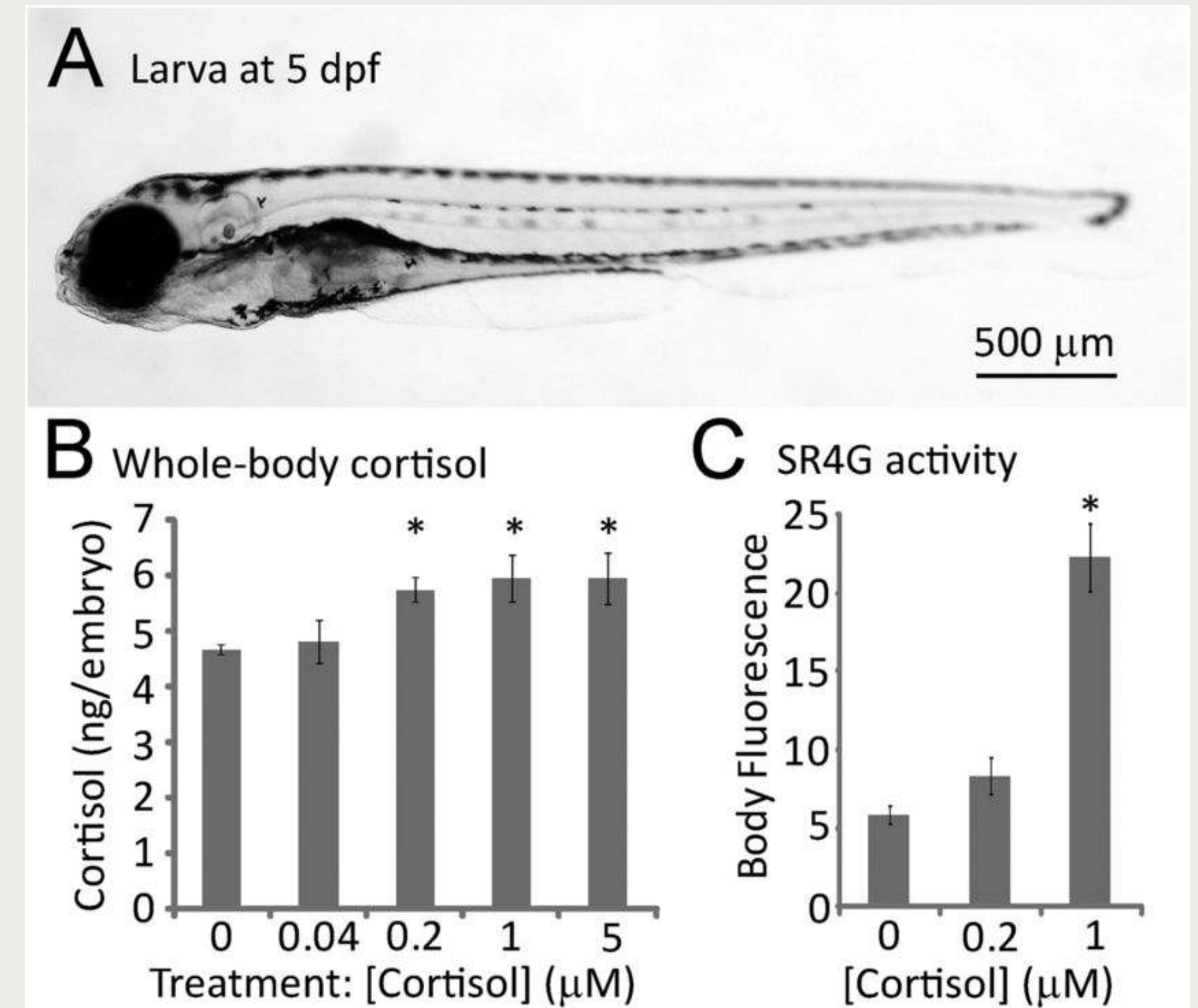
Weikum 2017, Nature Reviews Molecular Cell Biology

Why quantify mRNA expression?

Cortisol-treated zebrafish embryos develop into pro-inflammatory adults with aberrant immune gene regulation

Major research question:

How does chronically elevated cortisol during early development affect gene expression in adulthood?



Ways to measure mRNA quantity

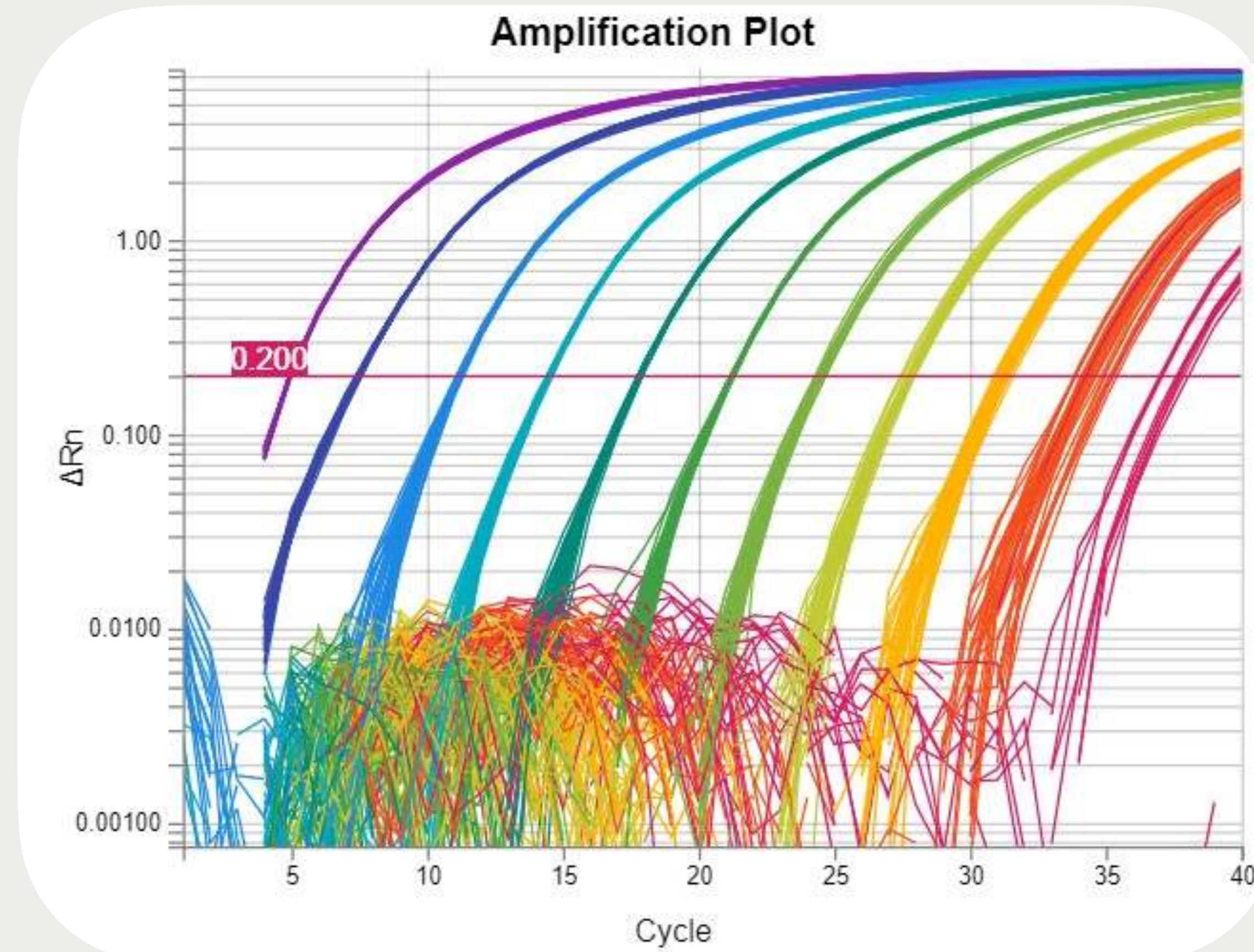
RT-qPCR: one to dozens transcripts at a time

Real-time quantitative polymerase chain reaction

1. Begin with RNA template (i.e., all RNA from an animal)
2. Convert to complementary DNA (**cDNA**) with **reverse transcriptase**
3. Amplify cDNA with Taq polymerase and **fluorescent dNTPs**
4. Thermocycler cycles through melt (denature), anneal, and extension temperatures
 - a. Sensor detects the fluorescence of intercalating DNA dye

Ways to measure mRNA quantity

RT-qPCR: one to dozens transcripts at a time



Drawbacks:

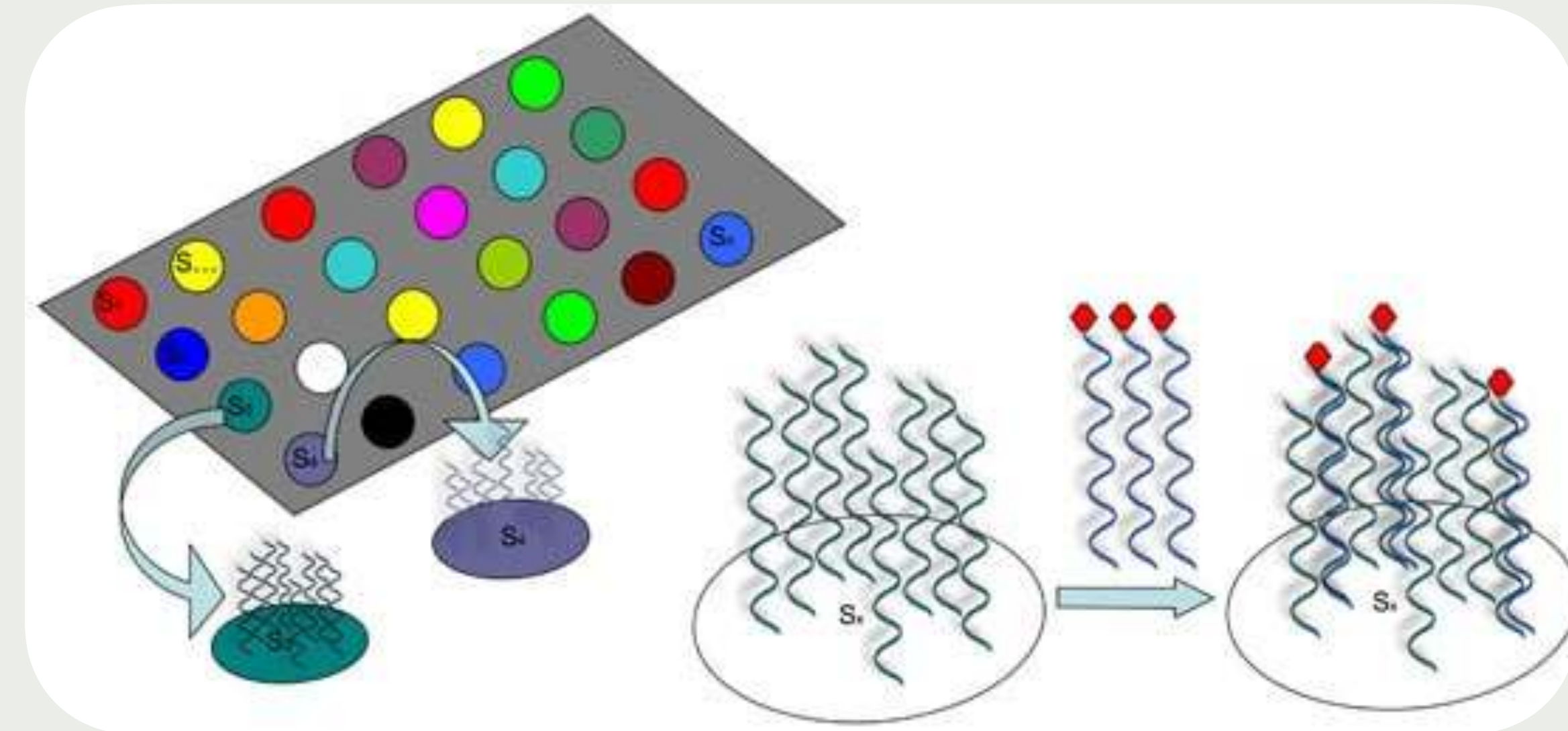
- 1) Required to design primers for each gene of interest*
- 2) Number of genes of interest limited by technology (thermocyclers, well plates)*

Ways to measure mRNA quantity

Microarray: hundreds to thousands of transcripts at a time

Microarray

1. Begin with RNA template (i.e., all RNA from an animal)
2. Convert to complementary DNA (**cDNA**) with **reverse transcriptase** and **fluorescent dNTPs**
3. Design a microchip with an array of probes
4. Melt cDNA and **hybridize** to the chip
5. Detect fluorescent spots on chip



Ways to measure mRNA quantity

RNA sequencing: unlimited number of transcripts

RNA-seq

1. Begin with RNA template (i.e., all RNA from an animal)
2. Convert to complementary DNA (**cDNA**) with **reverse transcriptase**
 - a. Optional: PCR amplify to increase sensitivity
3. Prepare **sequencing libraries**
4. Sequence

Drawbacks:

- 1) *Significant expertise needed for data quality control and analysis*

Ways to measure mRNA quantity

RNA sequencing: unlimited number of transcripts

